

JEE MAIN REVISION MASTER: LIQUID SOLUTION

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Raoult's Law & Colligative Properties	32 PYQs	Colligative Properties

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Raoult's Law & Colligative Properties	$\Delta T_b = i K_b m$, $\Delta T_f = i K_f m$, $\pi = i C R T$, $i = 1 + (n-1)\alpha$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Raoult's Law & Colligative Properties (Very High Yield — 32 PYQs)

Core Principle: Raoult's Law: $P = P_A^\circ \chi_A + P_B^\circ \chi_B$. Colligative properties depend on the number of solute particles: RLVP ($P^\circ - P / P^\circ$) = $i \chi_B$; boiling point elevation $\Delta T_b = i K_b m$; freezing point depression $\Delta T_f = i K_f m$; osmotic pressure $\pi = i C R T$. Van 't Hoff factor $i = 1 + (n-1)\alpha$ for dissociation; $i = 1 + (1/n - 1)\beta$ for association.

■ **Caution / JEE Pitfall:** Osmotic pressure is the BEST colligative property for determining molar masses of biomolecules and polymers because measurements are made at room temperature and produce large, easily readable values.

■ **Speed Trick:** A 0.1 M $Al_2(SO_4)_3$ solution has $i=5$, so it will produce the greatest depression in freezing point (lowest freezing point) among equimolar solutions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol⁻¹) in 100 g of water is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "When 0.0106 mole of acetic acid was dissolved in 1 kg of water, the freezing point depression for this strength of acid was 0.0205 K. ..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Match List-I (Laws) with the List-II (Mathematical expressions): List-I List-II (a) Henry's law (i) $p_1 = p_1^0 \chi_1$ (b) Raoult's law ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol^{-1}) in 100 g of water is

- (A) 0.025
- (B) 0.512
- (C) 0.051
- (D) 0.250

Correct Answer: None

Solution Analysis: For a non-volatile solute like urea, the relative lowering of vapour pressure is equal to the mole fraction of the solute. $\frac{p^\circ - p}{p^\circ} = x_{\text{(solute)}}$ Moles of urea: $n_{\text{(urea)}} = (18 / 60) = 0.3 \text{ mol}$ Moles of water: $n_{\text{(water)}} = (100 / 18) = 5.56 \text{ mol}$ Now, mole fraction of urea is $x_{\text{(urea)}} = \frac{n_{\text{(urea)}}}{n_{\text{(urea)}} + n_{\text{(water)}} = \frac{0.3}{0.3 + 5.56}$ $x_{\text{(urea)}} = (0.3 / 5.86) = 0.051$ So, the relative lowering of vapour pressure is 0.051 Correct option: Option C.